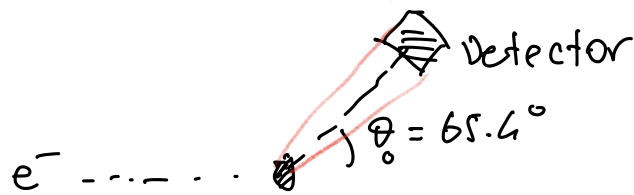


Electron - nucleus scattering

$$E = 495 \text{ MeV} \gg m_e$$

fixed angle experiment



$$E' = \frac{E}{1 + 2 \frac{E}{M} (1 - \cos \theta)}$$

$$M \approx A \cdot m_p \approx 12 \cdot \text{GeV}$$

$$E' \approx 480 \text{ MeV} \quad \text{elastic scattering.}$$

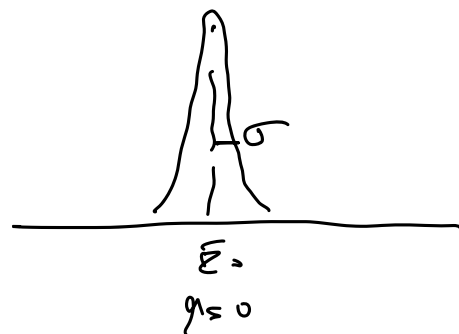
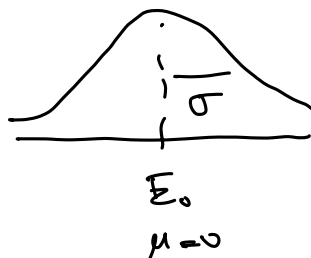
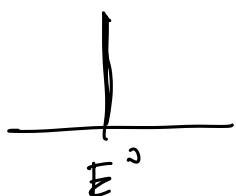


$$\# \text{ events} = \int_{\theta_0 - \Delta}^{\theta_0 + \Delta} \frac{d\sigma}{d\Omega} d\cos\theta d\varphi$$

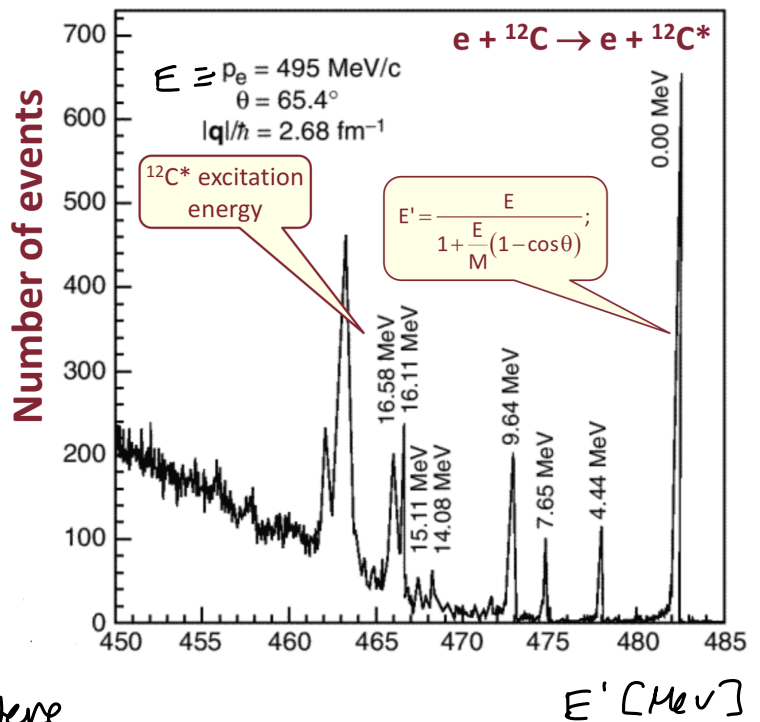
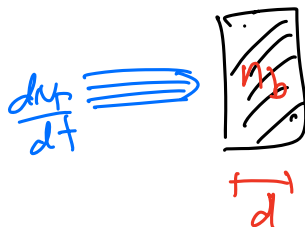
$$E' \ll E : \quad \text{inelastic scattering}$$

$$e + {}^{12}\text{C} \rightarrow e + {}^{12}\text{C}^*$$

$$\text{observed peak} \quad \int dE' \delta(E' - E_0) e^{-\frac{(E' - \mu)^2}{2\sigma^2}} \sim e^{-\frac{(E_0 - \mu)^2}{2\sigma^2}}$$



$$\frac{dN_r}{dt} = \sigma \cdot \frac{dN_p}{dt} \cdot n_b \cdot d$$



transition lines $\sim \delta(E')$
before detector resolution.

observed distribution
 $\delta(E') \otimes \text{Resolution}(E', \sigma)$

to explore structure of matter $\Rightarrow e^- + p \rightarrow ?$

1960 @ SLAC: Stanford Linear Accelerator Center
at different e^- energies

$$e^- + p \rightarrow e^- + p$$

$$e^- + p \rightarrow e^- + \Delta(1232)$$

$$e^- + p \rightarrow e^- + \pi^+ + n$$

$$e^- + p \rightarrow e^- + \pi^+ + \pi^-$$

$$e^- + p \rightarrow e^- + \pi^0$$

Increase $E_e \rightarrow$ few GeV's.

$m_{\text{target}} \rightarrow 1 \text{ GeV}$ (proton mass)

so for assumptions:

- pointlike $S=1/2$ probe (electron)
- pointlike spinless target (nucleus/proton)
heavy

Missing features:

- 1) recoil of target ($E_e \approx m_{\text{target}}$)
- 2) proton is not pointlike (structure of target)
- 3) spin of the target ($S=1/2$ for protons)

1) Recoil

$$\frac{d\sigma}{d\Omega} \propto \frac{\alpha^2}{p^4 \sin^4 \frac{\theta}{2}} \frac{|\vec{P}_{\text{out}}|}{|\vec{P}_{\text{in}}|} E^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

at low energy limit (Rutherford) $|\vec{P}_{\text{out}}| \approx |\vec{P}_{\text{in}}|$

$P_{\text{in}} \gg m_e \approx m_1$ (mass of probe).

$$P_{\text{in}} \approx E_{\text{in}} \approx E$$

$|\vec{P}_{\text{out}}| \approx E'$: energy of e^-
in final state

including recoil

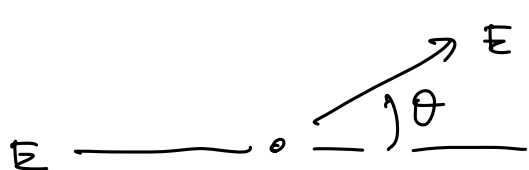
$$\frac{d\sigma}{d\Omega} \propto \frac{\alpha^2}{E^4 \sin^4 \frac{\theta}{2}} E^2 (1 - \beta^2 \sin^2 \frac{\theta}{2}) \frac{1}{1 + 2 \frac{E}{\mu} \sin^2 \frac{\theta}{2}}$$

Rutherford

Mott

includes recoil

Experimental observables: E, E', θ



2) structure of proton

$e^- \cdots \cdots$



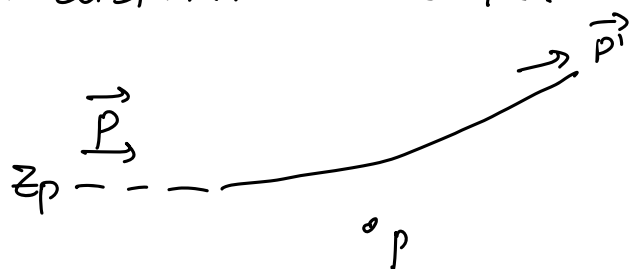
spatial distribution of charge.

$$\frac{d\sigma}{d\Omega} \propto |\langle f | H_I | i \rangle|^2 \underbrace{\rho(E)}_{\text{phase space}}$$

H_I charges for spatial distribution of charge.

point like particles:

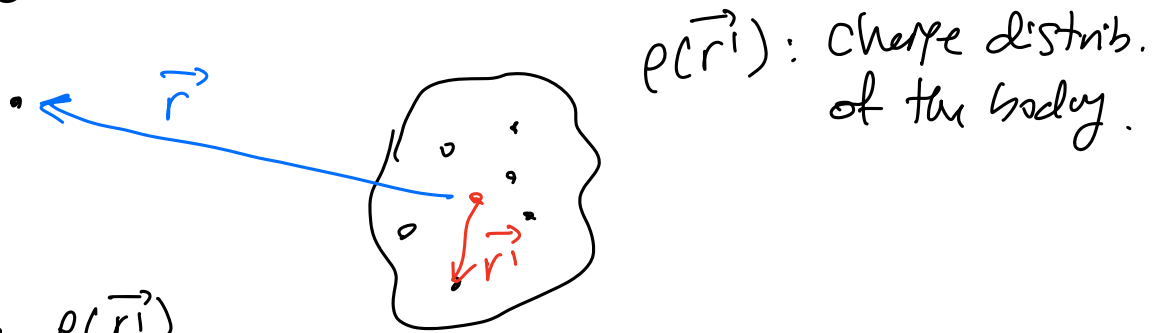
$$\psi(r) = \frac{1}{\sqrt{V}} e^{i \vec{p} \cdot \vec{r}}$$



$$\mathcal{M} \sim \int d^3r e^{-i \vec{p}' \cdot \vec{r}} \frac{1}{r} e^{i \vec{p} \cdot \vec{r}} = \int d^3r \frac{e^{i \vec{q} \cdot \vec{r}}}{r} \propto \frac{1}{q^2}$$

$$\vec{q} = \vec{p} - \vec{p}'$$

Extended body with charge distribution



$$V(\vec{r}) = \int d^3\vec{r}_i \frac{\rho(\vec{r}_i)}{|\vec{r} - \vec{r}_i|}$$

pointlike: $\rho(\vec{r}_i) = Z_T \cdot e \cdot \delta(\vec{r}_i)$

Extended body: $\rho(\vec{r}_i) = Z_T \cdot e \cdot f(\vec{r}_i)$

$$M = \langle f | V(r) | i \rangle \sim Z_p Z_T e^2 \int d^3\vec{r} e^{i\vec{q} \cdot \vec{r}} \underbrace{\int d^3\vec{r}_i \frac{f(\vec{r}_i)}{|\vec{r} - \vec{r}_i|}}_{V(r)}$$

$$M \sim \int d^3\vec{r} \frac{e^{i\vec{q} \cdot (\vec{r} - \vec{r}_i)}}{|\vec{r} - \vec{r}_i|} \underbrace{\int d^3\vec{r}_i e^{i\vec{q} \cdot \vec{r}_i} f(\vec{r}_i)}_{F(q^2) : \text{Form Factor}}$$

$F(q^2)$: Fourier transf. of spectral distribution.

$$M_{\text{extended body}} \sim \frac{Z_T Z_p e^2}{q^2} F(q^2)$$

$M_{\text{pointlike}}$

$$\frac{d\sigma}{d\Omega} \propto |M|^2 \Rightarrow \left. \frac{d\sigma}{d\Omega} \right|_{\text{extended body}} = \left. \frac{d\sigma}{d\Omega} \right|_{\text{pointlike}} \times |F(q^2)|^2$$

How to measure $F(q^2)$?

$$q^2 = ?$$

$$\underline{q} = \underline{p}_1 - \underline{p}_3$$

$$\underline{p}_1 = (\underline{E}, \underline{\vec{p}}_1) \quad \underline{p}_3 = (\underline{E}', \underline{\vec{p}}_3)$$

$\xrightarrow{\quad} \theta$

$$E \gg m_e$$

$$E \approx |\underline{\vec{p}}_1| \quad E' \approx |\underline{\vec{p}}_3|$$

$$q^2 = (\underline{p}_1 - \underline{p}_3)^2 = \underline{p}_1^2 + \underline{p}_3^2 - 2 \underline{p}_1 \cdot \underline{p}_3 =$$

$$= m_e^2 + m_e^2 - 2(\underline{E}' \underline{E} - \underline{\vec{p}}_1 \cdot \underline{\vec{p}}_3)$$

$$= 2m_e^2 - 2(E E' - E E' \cos \theta)$$

$$\approx -2 E E' (1 - \cos \theta)$$

$$\approx -4 E E' \sin^2 \frac{\theta}{2}$$

$$\underline{p}_1^2 = E^2 - |\underline{\vec{p}}_1|^2 = m_e^2$$

$$\underline{\vec{p}}_1 \cdot \underline{\vec{p}}_3 = E \cdot E' \cos \theta$$

$$m_e^2 \ll E E'$$

measure $F(q^2)$ as function of q^2

$q^2 \approx 0$ accessible with small θ angles.

Measure $F(q^2)$ for $q^2 \rightarrow 0$

$$F(q^2) = (\text{norm}) \iiint e^{i \underline{q} \cdot \underline{r} \cos \theta} f(r) r^2 dr d\cos \theta d\varphi$$

$$q \neq 0 \quad e^{i \underline{q} \cdot \underline{r} \cos \theta} = 1 + i \underline{q} \cdot \underline{r} \cos \theta - \frac{1}{2} (q r)^2 \cos^2 \theta + \mathcal{O}(q r)^3$$

$$\Rightarrow F(q^2) \approx 1 - \frac{1}{6} q^2 \langle r^2 \rangle$$

$$\langle r^2 \rangle = \int_0^\infty f(r) r^2 r^2 dr$$

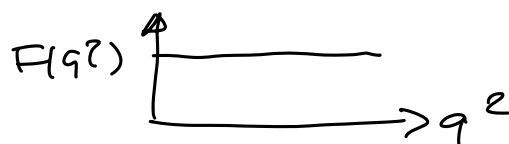
meaning of $\langle r^2 \rangle \neq 0$?

Examples of $f(r)$

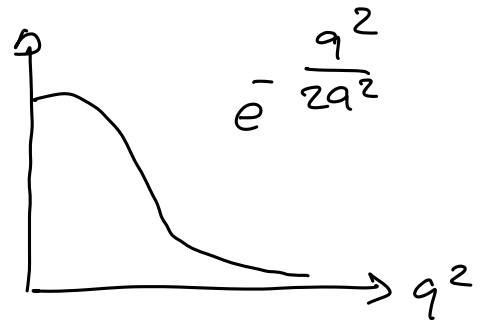
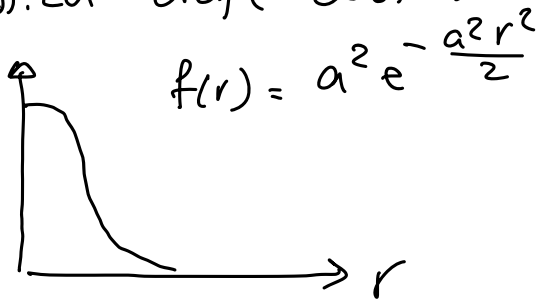
1) pointlike object



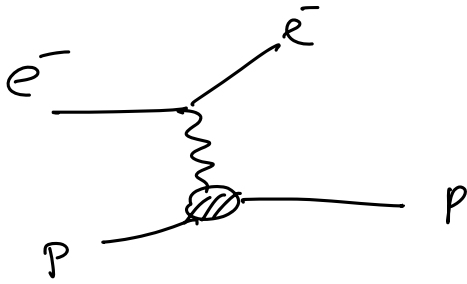
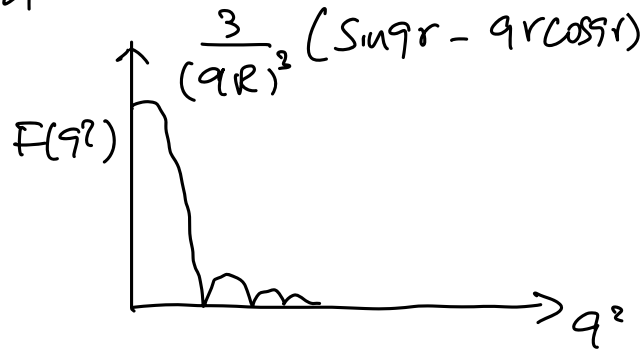
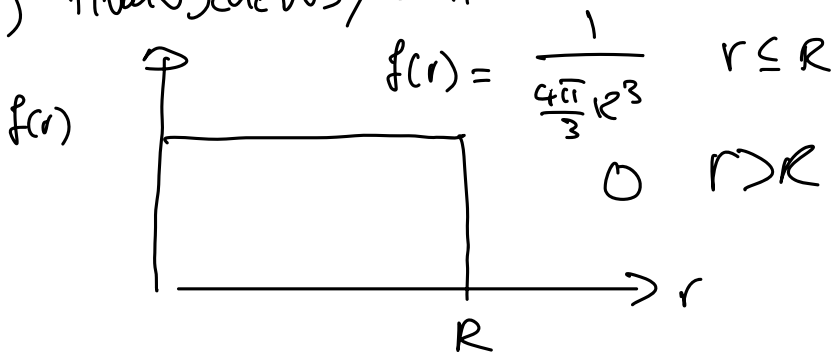
$$f(r) = \frac{e}{4\pi} \delta(r) \quad F(q^2) = 1$$



2) Gaussian charge distribution.



3) Homogeneous/uniform distribution



$$\frac{d\sigma}{dq^2} = (-\dots) \frac{\alpha^2}{q^4} E^2 (1 - \beta^2 \sin^2 \frac{\Theta}{2}) \frac{E'}{E} |F(q^2)|^2$$

$$q^2 = -4EE' \sin^2 \frac{\Theta}{2}$$

$$\left. \frac{d\sigma}{dq^2} \right|_{\text{structure}} = \left. \frac{d\sigma}{dq^2} \right|_{\text{point}} \times |F(q^2)|$$

$$R = \frac{\left. \frac{d\sigma}{dq^2} \right|_{\text{struct}}}{\left. \frac{d\sigma}{dq^2} \right|_{\text{point}}} = |F(q^2)| \approx 1 - \frac{1}{6} q^2 \langle r^2 \rangle$$

for $q^2 \rightarrow 0$.

measure from data $\rightarrow$ well known and computed.

$F(0) \approx 1 \Rightarrow$ point like

otherwise $\langle r^2 \rangle \neq 0 \Rightarrow$ structure

3) spin for target proton

magnetic moment $\mu = g \frac{e\hbar}{4m}$

electron $g = 2$ $\mu_e = \frac{e\hbar}{2m}$ $m_e = 0.511 \text{ MeV}$

proton $g = 2$ expected

neutron $g = 0$ expected

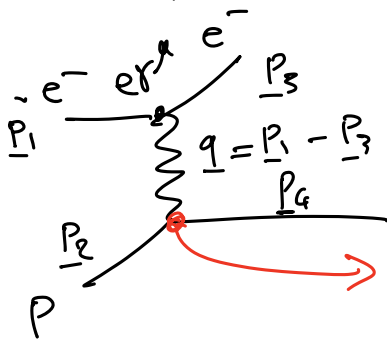
experimentally:

$g_p = 2.79$

$g_n = -1.91$

} anomalous magnetic moment

$\Rightarrow$ indication that are not elem. particles.



$F_1(q^2) \gamma^\nu + F_2(q^2) i \frac{\sigma^{\nu\tau} q_\tau}{2m_p} K$

currents

mag. mom. interaction.
(spin of target)

F_1, F_2 : are two unknown form factors

$K = g - 1$

F_1 : spatial charge distribution.

$\sigma^{\nu\tau} = \frac{i}{2} [\gamma^\nu, \gamma^\tau]$ γ^ν : Dirac matrices.

$F_1(q^2=0) = F_2(q^2=0) = 1$

$\mathcal{M} \sim e [\bar{u}(p_3) \gamma^\mu u(p_1)] \left[\frac{-i g^{\mu\nu}}{q^2} \right] e \bar{u}(p_4) [F_1 \gamma^\nu + \frac{q^\nu}{2} - F_2] u(p_2)$

$\frac{d\sigma}{d\Omega} \propto |\mathcal{M}|^2 \rho(E)$

$\left. \frac{d\sigma}{d\Omega} \right|_{\text{Rosenbluth}} = (-) \frac{d^2}{q^4} E^2 \underbrace{\cos^2 \frac{\theta}{2}}_{\beta=1} \frac{E'}{E} \left[\left(F_1^2 - \frac{K^2 q^2}{4M^2} F_2^2 \right) + (F_1 + KF_2)^2 \frac{(-q^2)}{2M^2} \tan^2 \frac{\theta}{2} \right]$

$$(\bar{F}_1^2 - \cancel{\frac{k^2 q^2}{4M^2}} \bar{F}_2^2) \cos^2 \frac{\theta}{2} + (\bar{F}_1 + k\bar{F}_2)^2 \cancel{\frac{(-q^2)}{2M^2}} \sin^2 \frac{\theta}{2}$$

$$q^2 = -4EF' \sin^2 \frac{\theta}{2}$$

$$M = 1 \text{ GeV}$$

$$q^2 \ll M^2 \Rightarrow \left. \frac{d\sigma}{d\Omega} \right|_{\text{Rosenbluth}}^N \frac{d\sigma}{d\Omega} \Big|_{\text{part}}^X \left| F_1(q^2) \right|^2$$

Experiment by McAllister, Hofstadter in 1956
Nobel prize 1961.

$$e^- + {}^4\text{He} \rightarrow e^- + X$$

$$E_e = 188 \text{ MeV}$$

measure E' vs θ

$$\theta \in [35^\circ, 138^\circ]$$

Goldhaber Ch. 8

$$\frac{E'}{E} = \frac{1}{1 + 2 \frac{E}{M} \sin^2 \frac{\theta}{2}}$$

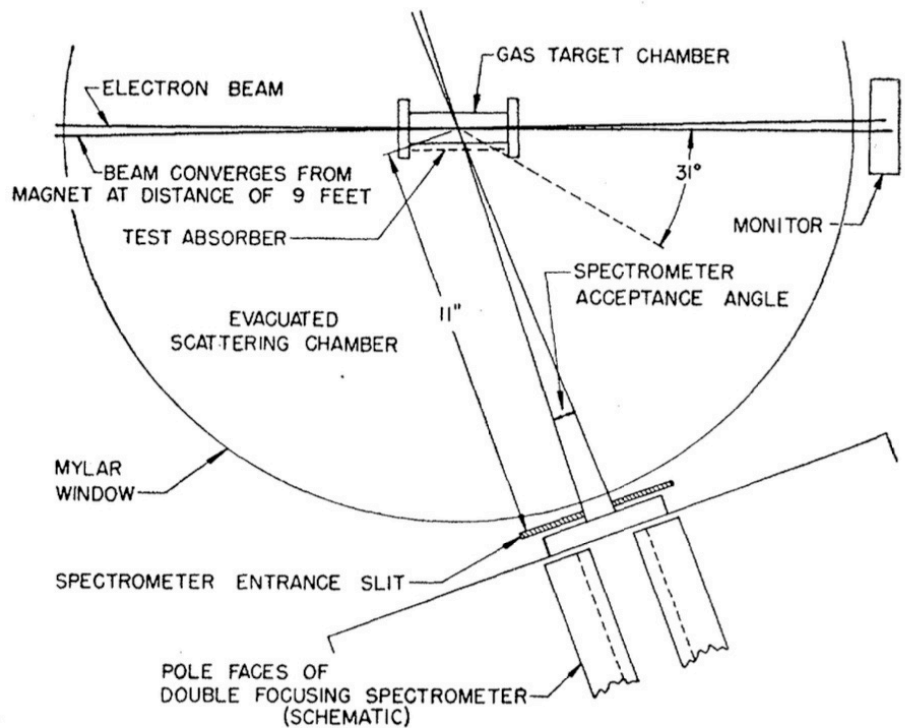
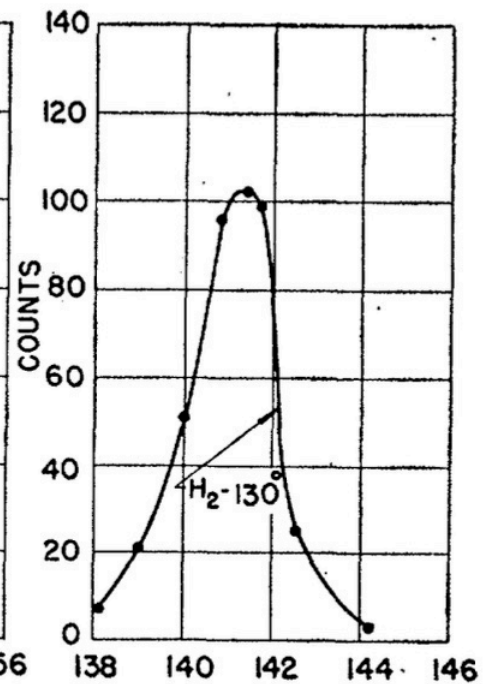
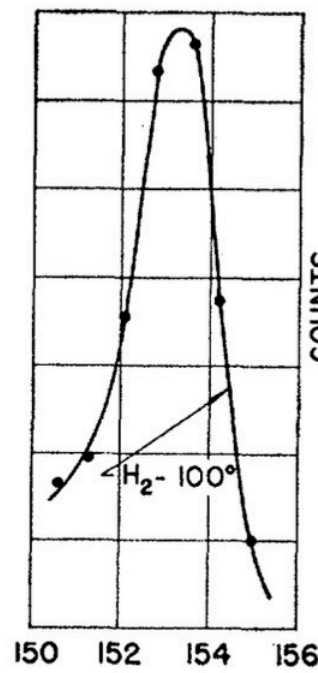
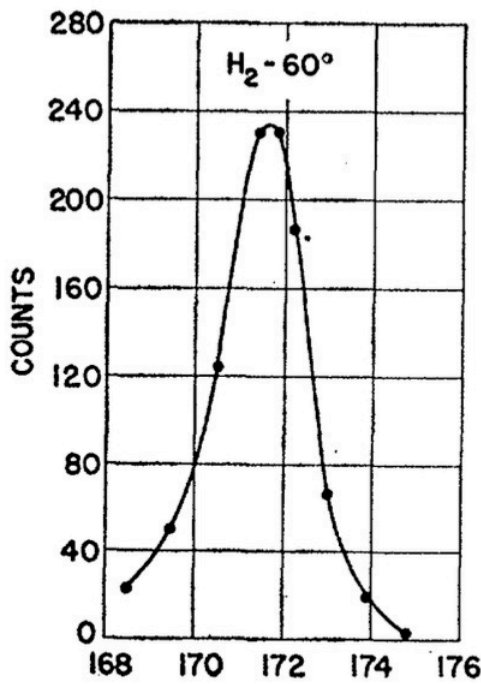


FIG. 2. Arrangement of parts in experiments on electron scattering from a gas target.



ENERGY IN MEV

$$\frac{d\sigma}{d\Omega} \propto \cos^2 \frac{\theta}{2} \Rightarrow \# \text{events vs. } \theta$$

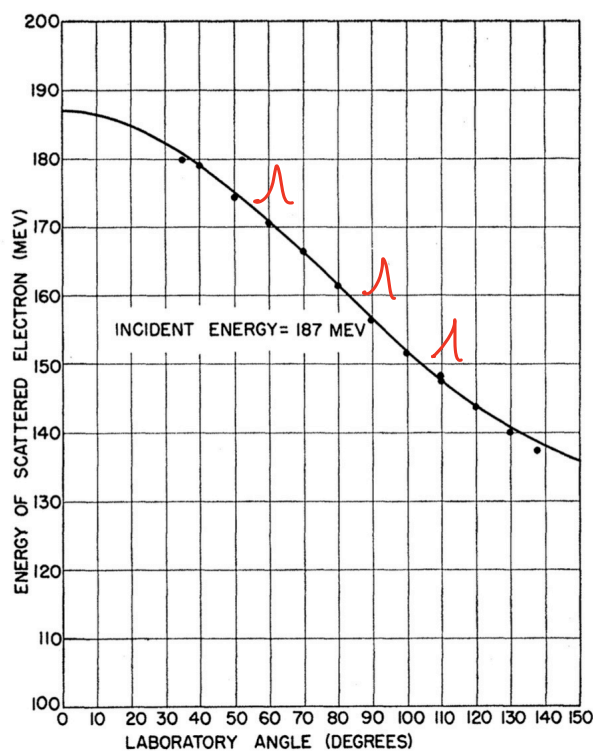
$$E' = \frac{E}{1 + 2 \frac{E}{\mu} \sin^2 \frac{\theta}{2}}$$

$$\theta = 60^\circ \quad \frac{\theta}{2} = 30^\circ$$

$$\sin 30^\circ = \frac{1}{2}$$

$$E = 188 \text{ MeV}$$

$$E' \approx 175 \text{ MeV}$$



Data reproduces
expected E'/E